

Mean State of The Tropical Atmosphere

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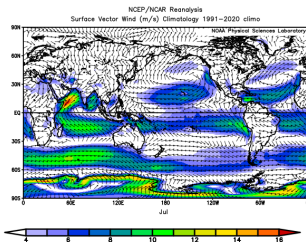
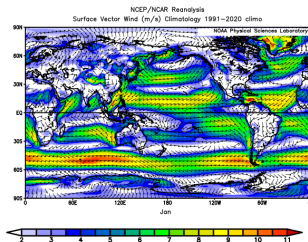
MEA 458/NC State University

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- In the following slides, we will examine the winter (January) and Summer (July) mean flow.
- Record your observations as we examine each slide

Surface Winds

Surface Winds: January Mean and July Mean

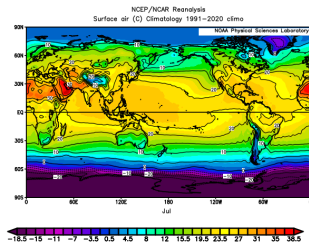
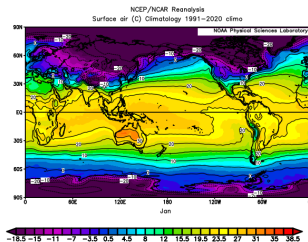


Key observations:

- Equatorward flowing Trade winds in both summer and winter season over ocean basins in both hemispheres
- **Exception:** In the North Indian Ocean (Arabian Sea and Bay of Bengal), the trade winds are replaced by monsoon southerlies in Summer

Surface Temperature

Surface Temperature (C): January Mean and July Mean

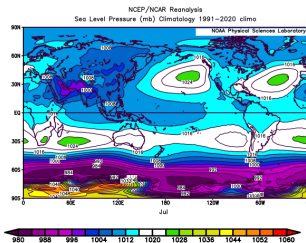
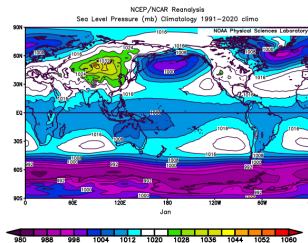


Key observations:

- Weak temperature gradients in the tropics and relatively stronger gradients in the mid-latitudes.
- Continental hot-spots during summer in each hemisphere. During Jan, note the heating maxima (heat-lows) over South Africa and Australia; and during Jul, over North Africa, middle-east, North India, Southwestern US.

Sea Level Pressure

Sea Level Pressure: January Mean and July Mean

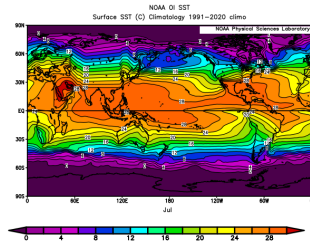
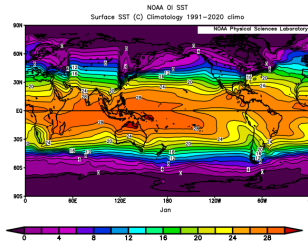


Key observations:

- Weak pressure gradients in the tropics and relatively stronger gradients in the mid-latitudes.
- Subtropical anticyclones over the oceans. They are stronger and more expansive in the mean during summer. Also note how they are closer to the continent to their west. We will examine this more in the modules on Hadley Circulation and Monsoons.

Sea Surface Temperature

SST: January Mean and July Mean

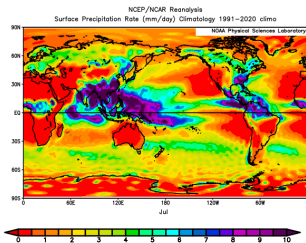
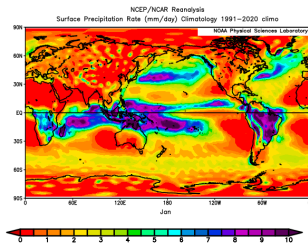


Key observations:

- Warmest waters over eastern Indian and western Pacific (the Indo-Pacific warm pool). The eastern Pacific is relatively cooler.
- During strong El Nino, the warm pool expands towards the central and eastern Pacific.
- Again weak gradients in the tropics compared to mid-latitudes.

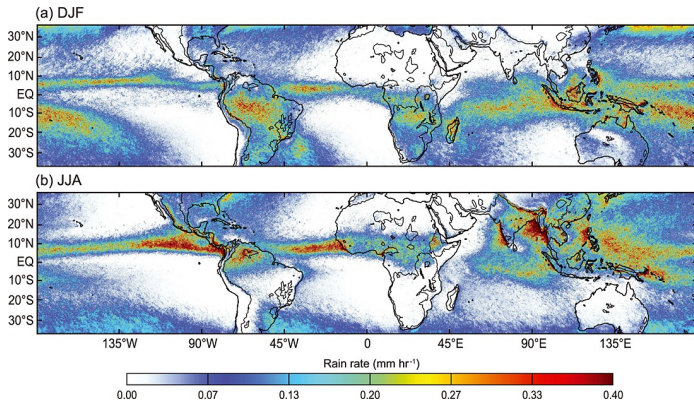
Precipitation

Precipitation: January Mean and July Mean



Precipitation

Precipitation: Dec-Feb Mean and June-Aug Mean



Source: Houze et al. (2015); uses NASA TRMM Satellite derived estimates

Key observations:

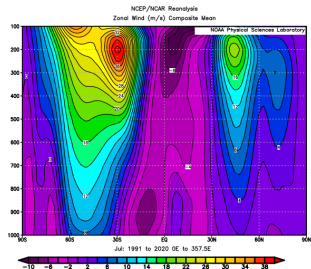
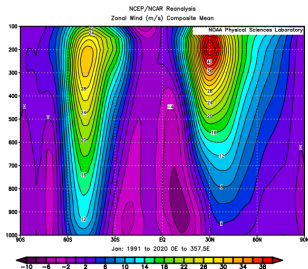
- In the tropics, precipitation seems to generally (but not entirely) coincide with the warmest ocean
- Clearly precipitation is concentrated in a narrow band near the equator over the Pacific and Atlantic (The ITCZ)
- ITCZ less well defined over the western Pacific and Indian Ocean - here there is more widespread precipitation
- Note the monsoon rainfall over south Asia during the summer month, with peaks over western coast of India. foothills of the Himalayas, and in the Bay of Bengal

Key observations (contd.):

- Note a band of rainfall extending southeastward in the western Pacific. This is the South Pacific convergence zone (SPCZ)
- Clear lack of precipitation in the subtropics (ocean as well as land), the desert regions of the world.

Zonal Wind - Vertical Section

Zonal Wind: January Mean and July Mean



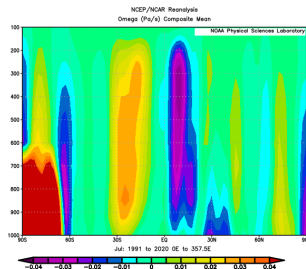
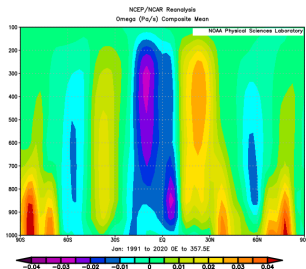
Zonal Wind - Vertical Section

Key observations:

- Upper-level subtropical westerly jets seen in both hemispheres. Stronger and more equatorward during winter. Thermal wind balance is a key dynamical explanation for these jets
- Near surface easterlies over much of the tropics (the trade winds).
- A strong upper-level easterly jet just to the north of the equator is seen in summer. This is, in part, reflects the tropical easterly jet and is related to the summer Asian monsoon.
- Fun fact: near surface westerlies speed up the earth's rotation, while easterlies slow it down. check out How long was yesterday?

Omega - Vertical Section

Vertical Velocity (ω): January Mean and July Mean



negative values $\rightarrow$ ascent; positive values $\rightarrow$ descent

Key observations:

- We see the Hadley circulation
- Ascent near the equator and descent in the subtropics
- Winter and summer Hadley cells are not symmetric.
- More on this in the Hadley circulation module.